Full Name: Ahmad Jamal Abuadas
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E-mail Address: ahmadabuadas2013@gmail.com **Data Engineer\Big Data**

Data Engineer || ETL || Snowflake ||Pyhton ||Cloud (AWS) || Hadoop || Databricks || Business intelligence Developer || Data Visualizations || Data Analyst || SQL || Big Data Tools || ML || AI GEN AI

Data Engineer with over a decade of experience in software engineering, I've spent the **last 7 years specializing in Data and Big Data Engineering**. I completed my studies with a **master's in information technology**, specializing in Big Data. This academic journey provided me with the theoretical knowledge that pairs well with my hands-on experience in the field. My proficiency in Python, SQL, Pyspark, and extensive experience with cutting-edge technologies such as Hadoop, Hive, Hue, Shell Scripting, Apache Spark, Cloudera, and AWS(S3, EC2, Redshift, IAM, ..) positions me as a skilled and innovative data engineer.

Specialized in **Big Data ecosystem,Ingestion, Modeling, Storage Analysis, Integration,** and **Data Processing,** development, implementation, deployment, and maintenance using Python, Hadoop and Spark related technologies using **Cloudera, CDP**, Amazon EMR.

PROFESSIONAL PROFILE

- Created and optimized **PySpark logic** for complex data transformations, significantly improving efficiency and accuracy in reconciliation processes.
- Employed **SQL for data querying and manipulation, ensuring data quality** and adherence to regulatory standards.
- Support production issue for ETL/ELT jobs, spark jobs, database issues Debugging and fix issues.
- Utilized automation tools such as **Autosys, Jenkins, and RLM** for scheduling and orchestrating data jobs, resulting in improved job reliability and minimized downtime.
- Successfully integrated Continuous Delivery/Continuous Integration **(CDCI)** practices into the automation framework, ensuring seamless development, testing, and deployment of data solutions.
- Played a crucial role in ensuring the stability and scalability **of reconciliation processes**, allowing the business to adapt to evolving market conditions.
- Setting up **AWS EC2, S3, IAM within the Kelvin platform** testing process Designed bridges to connect with devices for data transmission using Python scripts.
- Developed POV UI integrated with the Kelvin platform **using Python (Streamlit)**
- Collaborated with the product team to create Kelvin Maps by set up control system OPC configuration to extract data, connecting it **to OPC Bridge/MQTT** Bridge for integration into the Kelvin platform.
- Develop and maintain **Java code** for risk management system collaborated with the product team
- Developed EDHR and RMT system real-time data-driven applications using PLC code, DeviseWise, and the Oracle database and support Java web applications, showcasing proficiency in full-stack development, user interface design, and backend logic.
- Leveraged Snowflake as a robust Data Warehouse solution, capitalizing on its scalability, performance, and flexibility to handle extensive datasets efficiently.

- Developed Alarm and event tracking system, a real-time data-driven application extracting data from the **SQL database using Python**.
- Developed solutions for reconciliation projects, ensuring data accuracy and consistency across all markets in different regions.
- Developed a Spare Part Management System using **Ruby on Rails and JS**.
- Utilized **PLC coding and HMI** applications to store sensitive data, meeting business requirements.
- Utilized **Databricks to run schedule jobs flow for sql query & pyspark jobs** use ingestion tool at **Databricks to connect with S3 and automated triggers for job flow.**
- Utilized HADOOP, Spark, and Scala to implement robust data processing solutions, enhancing system performance and scalability.
- Create ETL jobs to load and mange the data from logs, systems, ODBC in to Mongo and Postgres databases.
- Build reports using Tableau, RShiny, Streamlit, SSRS, Excel.
- development ETL pipline and loading data in to Mongo and Postgres databases as backend for web applications.

TECHNICAL SKILLS

Hadoop Components / Big Data	HDFS, Hue, MapReduce, Pig, Hive, Hbase, Sqoop, Impala, Zookeeper, Kafka, Yarn, Cloudera Manager, ,pyspark Airflow, Kafka , Snowflake, Scala
Languages	Python, R, Ruby , JS,C#, Java, SQL, Hive
IDE Tools	Visual Studio Code, Jupyter lab, Rstudio
Cloud platform	AWS (Lambda,Redshift , S3, EC2, EMR, RDS, AWS Glue , Kinesis,IAM,)Databricks
Reporting Tools	Tableau, RShiny, Streamlit, SSRS, Excel
Databases	MSQLServer,MySQL,Postgres ,Oracle, NoSQL Database (MongoDB , Hbase, Cassandra)
Data Analysis	Pandas, NumPy, SciPy, Scikit-learn, RShiny, streamlit, Plotly, Matplotlib,
CI/CD Tools	Jenkins, RLM, Bitbucket, Autosys, Airflow,Docker, Kubernetes,Helm
Software Methodologies	Agile, Scrum, Waterfall
Cloud platform	AWS , GCP, Docker, Kubernetes,Helm
Big Data Ecosystem	MapReduce, HDFS, HBase, Spark, Scala, Zookeeper, Hive, Pig, Sqoop Cassandra, Oozie, MongoDB, Flume, Databricks
ETL Tools	**Alteryx, DeviceWise, Talend, Airflow, DBT**
ML, AI, GEN AI	Generative AI development, **LangChain**, **OpenAI, LLMs**, vector databases, prompt engineering, and agent-based architectures, **RAG**
Programming Languages	Java, Python(Tensoflow, PyTorch, Keras, Numpy, SCipy, NLTK, Gensim, SpaCy,Pasndas, Matplotlib, Plotly), Linux shell scripts, R(Tidy, ggplot, tidyverse, Shiny).
Visual diagrams tools	Visio, Miro , Notaion

other	Support production , fixing bugs, check logs, tacking tickets

CAREER SUMMARY / Professional Experience

Citi Group, FL **SEP 2022 - Present**

Big Data Engineer

Project:

As a Big Data Engineer at Citi Group (Sep 2022 - Current), I have effectively harnessed cutting-edge Big Data tools and technologies, including Hadoop Distributed File System (HDFS), Apache Spark, Apache Hive, and Hue. My role involves processing and analyzing large-scale financial data efficiently. I am adept at collaborating with cross-functional teams, understanding project requirements, and formulating data engineering strategies to meet business objectives.

Responsibilities:

- **Built Generative AI Solutions using LangChain, OpenAI, and ML Frameworks**
  Developed intelligent AI agents and LLM-powered workflows for real-world automation tasks; integrated with vector databases and orchestrated via LangChain.
- **End-to-End Data Platform Migration from Hive to Snowflake**
  Designed and executed a large-scale cloud data migration strategy including schema conversion, data pipeline refactoring, performance optimization, and validation.
- **Engineered Advanced Solutions on Snowflake with Task Automation, Data Quality, and Unit Testing**
  Implemented Snowflake-native automation using **Tasks**, **Streams**, and **Procedures**; developed reusable **data quality checks** and **unit testing** frameworks using dbt and SQL.
- **Built AI-Powered Agents with LangChain and OpenAI for Data and Business Automation**
  Created domain-specific autonomous agents capable of task planning, decision-making, and tool use by leveraging LangChain's agent framework and OpenAI APIs.
- **Developed Interactive Dashboards and Monitoring Solutions**
  Created real-time monitoring dashboards using tools like **Streamlit**, **Power BI**, or **Looker** to visualize pipeline health, DQ metrics, and AI model outputs.
- Leveraged cutting-edge Big Data tools and technologies, including Hadoop Distributed File System (HDFS), Apache Spark, Apache Hive, and Hue, to process and analyze large-scale financial data efficiently.
- Collaborated with cross-functional teams to understand project requirements and formulated data engineering strategies to meet business objectives.
- Created and optimized PySpark logic to perform complex data transformations and calculations, resulting in improved efficiency and accuracy in reconciliation processes.
- Employed SQL for data querying and manipulation, ensuring data quality and adherence to regulatory standards.
- Utilized automation tools such as Autosys, Jenkins, and RLM for scheduling and orchestrating data jobs, improving job reliability and minimizing downtime.
- Developed shell scripts to automate routine tasks, enhance system performance, and streamline data processes.

- Integrated Continuous Delivery/Continuous Integration (CDCI) practices into the automation framework, ensuring seamless development, testing, and deployment of data solutions. - Successfully delivered high-performance data solutions, reducing reconciliation discrepancies and enhancing overall data quality.
- Played a crucial role in ensuring the stability and scalability of the reconciliation processes, allowing the business to adapt to evolving market conditions.
- Used PySpark to write the code for all the use cases in spark and extensive experience with Scala for data analytics on Spark cluster and Performed map-side joins on RDD.
- Involved in converting Hive/SQL queries into Spark transformations using Spark RDDs, Scala and have a good experience in using Spark-Shell .
- Responsible for estimating the cluster size, monitoring and troubleshooting of the Spark jobs on clodera CDP. Support production issue for ETL/ELT jobs, spark jobs, database issues Debugging and fix issues.
- Experience with Snowflake cloud data warehouse and AWS S3 bucket for integrating data from multiple source system which include loading nested JSON formatted data into snowflake table.
- Developed custom the Unix/BASH **SHELL** scripts for pre and post validations of the **master** and **slave** nodes, before and after the configuration of the name node and data nodes
- Used Apache Airflow to manage the system workflows.
- Developed shell scripts and optimize it to run many jobs UNIX.
- Loaded the data from Teradata to HDFS using Teradata Hadoop connectors.
- Responsible to provide the technical solutions for the team facing issues.
- Responsible to guide the team when they have issues.
- Responsible to provide design and architecture to the team to develop applications.
- Creating data model for the data to be ingested for each table.
- Identifying the appropriate file formats for the tables to retrieve the data faster.
- Decided the column data types properly so that we do not lose the data or miss the data.
- Write Documentation and test case scenarios QA.

Kelvin INC ,CA **Mach 2022 – SEP 2022**
IT\OT Data Engineer:
Project Description:

In my role as an OT/IT Engineer at Kelvin INC, CA (2022 - 2022), I demonstrated expertise in setting up AWS EC2 within the Kelvin platform. Additionally, I designed bridges to connect with devices for data transmission using Python, developed POV UI integrated with the Kelvin platform using Python (streamlit), and collaborated with the product team to create Kelvin Maps. I set up control system OPC configuration to extract data, connecting it to OPC Bridge/MQTT Bridge for integration into the Kelvin platform. Employed advanced techniques to optimize data reading processes within Snowflake, ensuring swift and efficient retrieval of information for analytical purposes.

Responsibilities:

- Set up AWS EC2,S3, IAM within the kelvin platform for test env.
- Design bridges to connect with devices in sending data to the Kelvin platform using python.
- Discover use cases and develop POV UI integrated with the kelvin platform using python(streamlit)
- Working with the product team to develop Kelvin Maps.

- Set up control system OPC configuration to extract the data and connected it to OPC Bridge\MQTT Bridge in order to use it in the Kelvin platform.
- Developed job workflows in Oozie for automating the tasks of loading the data into HDFS.
- Collaborated with development and QA teams to understand API specifications and requirements.
- Utilized tools like Postman and RestAssured for API test automation, reducing testing time and enhancing efficiency.
- Create pipeline and config for Mongo DB database and modeling the collections to be used from web applications
- Use Airflow to schedule and monitoring the jobs
- Ingested the data from platform cloud storage in to
- Conducted security testing on APIs, identifying and addressing vulnerabilities to ensure data integrity.
- Worked closely with cross-functional teams to troubleshoot and resolve API-related issues.
- Integrated API testing into the CI/CD pipeline, automating regression tests for continuous integration.
- Designed and Implement test environment on AWS.
- AWS Snowflake, generating separate virtual data warehouses with differently sized classes.
- Developed scripts in Python (Pandas, Numpy,streamlit) for data ingestion, analyzing and data cleaning and Data sources are extracted, transformed and loaded to generate CSV data files.

Johnson & Johnson vision Health care, FL **APR 2019– APR 2022**
Software Data Engineer
Project Description:

During my tenure as a Software Data Engineer at Johnson & Johnson Vision Health Care I developed a data-driven application using Python, conducted SQL query writing, and utilized R studio to extract processes and visualize data. Established practices for ongoing enhancement and maintenance of the Snowflake Data Warehouse, focusing on optimizing query performance, refining data structures, and implementing updates to align with evolving business requirements. I implemented solutions and designed real-time data software such as EDHR, RMT, and YAS. My responsibilities included setting up data transaction triggers using PLC, DeviseWise, and the Oracle database. Additionally, I designed a fully managed ETL process using Python, R, DeviseWise, and PLC code.where I developed pipelines to extract the data from different sources, transform, and ingest using MQTT.

EDHR and RMT System Development:

Engineered real-time data-driven applications using PLC code, DeviseWise, and the Oracle database.Implemented innovative solutions for Electronic Device History Record (EDHR) and Raw Material Tracking (RMT) systems, ensuring streamlined data processing and accessibility. support Java web applications, user interface design, and backend logic.

YAS Application Development:

Pioneered the development of the Yield Analyses System (YAS), a real-time data-driven application.Extracted data from the vision system using Python and applied advanced ML and Text mining algorithms for in-depth data analysis.Visualized analytical insights using R code, enhancing decision-making processes through comprehensive data representation.

Alarm and Event Tracking System:

Developed an advanced Alarm and Event Tracking System, a real-time data-driven application.Extracted data from the SQL database using Python, creating a dynamic system for monitoring and responding to critical events promptly.

Backend for Analysis web applications:

Create ETL jobs to load and mange the data from logs, systems, ODBC in to Mongo and Postgres databases to make it ready as backend databases for web developing and data visualization teams.

Big data Analysis application

extracting data from diverse devices and orchestrating its ingestion into Hadoop HDFS. Leveraged PySpark for large-scale data processing and Scala, cleaning, and ML model development, predicting defects and reject numbers. Implemented workflow automation using Apache Airflow, ensuring seamless execution of data pipelines and ML workflows. Worked with Google Cloud Platform (GCP) nodes to deploy and manage scalable and reliable cloud infrastructure.

Responsibilities:

- Developing a data-driven application using Python, SQL query writing, and R studio to extract processes and visualize the data.
- Collaborated with cross-functional teams to design, implement, and optimize GCP node configurations for specific project requirements.
- Implements solutions and performs design for many real-time data software such as EDHR(Electronic Device History Record), RMT(Raw material tracking), and YAS(Yield Analyses System) by setup the data transaction triggers using PLC, DeviseWise, and Oracle database.
- Implemented streamlined ETL processes, integrating data seamlessly into Snowflake from various sources. This involved utilizing Snowflake's features for data transformation and loading, ensuring efficiency in data movement and transformation.
- **Read data from historian server and loaded in to OLAP Postgres database for analysis development and maintenance of data management systems, with a primary focus on.**
- Design fully managed ETL process using python, R, Devise wise, and PLC code.
- Use Talend for for data integration and management.
- Working across multiple environments (Dev, QA, UAT, Prod, etc.) to generate and check log files, and set up monitoring tools.
- **Load and read data for Mongo DB and use it as Backend for Flask API web application.**
- Design the ETL process using SQL on Snowflake, python, DeviseWise, and Alteryx.
- Developing automation Rockwell PLC software for 5th generation of contact lenses including Developing SCADA HMI (Human-machine interface) software.
- Developed logical & physical data model using data warehouse methodologies, including Star schema - Star-joined schemas, conformed dimensions' data architecture, early/late binding techniques, data modeling, designing & developing

- Worked on design and development of Informatica mappings, workflows to load data into staging area, data warehouse and data marts in SQLServer and Oracle
- Setup Enterprise network architecture, dataflow, and data layer drawing (using Visio)
- Used PySpark to write the code for all the use cases in spark and extensive experience with Scala for data analytics on Spark cluster and Performed map-side joins on RDD.
- Implemented Core Framework leveraging Spark that can handle pipeline in one Config.
- Designed and Implement AWS test use redshift as end datawarehouse .
- Engineered ETL workflows to seamlessly load and manage data from logs, systems, and ODBC into MongoDB, establishing a robust backend for web development initiatives.
- Implemented ETL processes to efficiently transfer and manage data from the same sources into PostgreSQL databases, catering to the specific backend needs of data visualization teams.
- Develop and maintain Java code for risk management system collaborated with the product team
- Build reports using Tableau, RShiny, Streamlit, SSRS, Excel.
- orchestrating the extraction of diverse device data into Hadoop HDFS and utilizing PySpark, Scala.

Fibertex Non-Woven, IL **Aug 2016– Oct 2018**
Software Data Engineer

Project Description:

Development of a Spare Part Management system, utilizing a tech stack comprising SQL database, Ruby on Rails, and JavaScript, ensuring seamless maintenance and troubleshooting of Control systems. Demonstrated hands-on expertise in managing PLC controllers from AB, Siemens, and other components like Simatic step7, Total Integrated Automation (TIA), RS Logix 5000. Proficiently utilized Rockwell automation mentoring software, including Studio5000, FactoryTalk, HMI client, Diagnostics Viewer, FactoryTalk Admin console, and OPC server. Played a pivotal role in enhancing safety systems by implementing IQ for safety PLCs, coupled with designing a user interface on the HMI to display safety element statuses, thereby significantly improving overall control system safety. Additionally, contributed to the development of an Axle Load Desktop App in civil engineering, utilizing VisualBasic, SQL Server, and Visual Studio to meet specific project requirements.

Responsibilities:

- Developing a data-driven application using Python, SQL query writing, and R studio to extract processes and visualize the data.
- Spearheaded the development of a sophisticated Spare Part Management system using a robust tech stack, including SQL database, Ruby on Rails, and JavaScript, ensuring optimal functionality and reliability.
- Applied in-depth knowledge in maintaining and troubleshooting Control systems, overseeing PLC controllers from industry leaders such as AB, Siemens, and various components like Simatic step7, Total Integrated Automation (TIA), RS Logix 5000.
- Demonstrated proficiency in utilizing Rockwell automation mentoring software, encompassing Studio5000, FactoryTalk, HMI client, Diagnostics Viewer, FactoryTalk Admin console, and OPC server, contributing to streamlined operations.

- Played a pivotal role in fortifying safety systems by implementing IQ for safety PLCs, and designing a user-friendly interface on the HMI, significantly enhancing the overall safety of control systems.
- Engineered a Spare Part Management System utilizing Ruby on Rails and JavaScript, showcasing adaptability with diverse technologies for comprehensive solutions.
- Employed PLC coding and HMI applications to securely store sensitive data, aligning with stringent business requirements for data integrity and confidentiality.

Public Works and Housing, Jordan **Feb 2012– March 2016**
Junior Engineer

Project Description:

This role involves a comprehensive evaluation of the technical specifications for the ministry's software systems, ensuring they meet the required standards and functionality. Collaborating closely with contractors, the individual contributes to the design and implementation of the Truck Weighing Scale System, overseeing both hardware installation and rigorous software testing to guarantee optimal performance. Additionally, they play a crucial role in validating schematic designs, collaborating with hardware engineers to ensure alignment with project objectives. Furthermore, the position entails a meticulous study of the technical offers presented by contractors and suppliers, actively participating in the acquisition process by obtaining referral procedures for tenders. Through these multifaceted responsibilities, the individual contributes to the seamless integration and enhancement of critical systems within the ministry's operational framework.

- Testing the technical specification of the ministry software systems.
- work with the contractors to design and implement the Truck weighing Scale System, including the hardware installation and software testing.
- Validated schematic designs working alongside hardware engineers.
- Studying the technical offers provides by the contractors and suppliers in obtaining referral procedures for tenders.